

IMAT REPAIR DRILL — Repeat Offenders

70 questions · 100 minutes · scoring: +1.5 correct, −0.4 wrong, 0 blank · Repair drill · answer + worked solution follow every question

Each question is immediately followed by its answer and worked solution, so this paper is for review and paper study rather than blind simulation. Letters refer to the option order printed here; the online app shuffles options on every attempt.

Section A - Respiration & Bioenergetics (Q1-12)

1. In a eukaryotic cell, the enzymes that catalyse the citric acid (Krebs) cycle are located mainly in the:

- A) intermembrane space
- B) cytosol
- C) mitochondrial matrix
- D) inner mitochondrial membrane
- E) outer mitochondrial membrane

Answer: C

The Krebs cycle enzymes sit free in the mitochondrial matrix, where acetyl-CoA enters the cycle (succinate dehydrogenase is the one exception, bound to the inner membrane). Placing the cycle in the inner membrane confuses it with the electron transport chain and ATP synthase, which are oxidative phosphorylation machinery, and glycolysis rather than the cycle runs in the cytosol.

2. Which of the following processes occurs in the mitochondrial matrix?

- A) reduction of O₂ to water at complex IV
- B) electron transfer between respiratory complexes I and IV
- C) conversion of glucose to pyruvate
- D) ATP synthesis driven by a proton gradient
- E) oxidation of acetyl-CoA to CO₂ in the citric acid cycle

Answer: E

Oxidising acetyl-CoA to CO₂ in the citric acid cycle happens in the matrix, where the cycle's enzymes sit. ATP synthesis driven by the proton gradient, electron transfer between complexes I and IV, and the reduction of O₂ to water all occur at the inner mitochondrial membrane, while conversion of glucose to pyruvate is glycolysis and happens in the cytosol. The recurring error is putting the whole respiratory chain inside the matrix.

3. In human cells, ATP synthase is located in the:

- A) outer mitochondrial membrane
- B) intermembrane space
- C) cytosol
- D) inner mitochondrial membrane
- E) mitochondrial matrix

Answer: D

ATP synthase is embedded in the inner mitochondrial membrane, with its catalytic head facing the matrix. It is driven by the H⁺ gradient between the intermembrane space and the matrix that the electron transport chain builds. Saying it sits free in the matrix confuses the site of the Krebs cycle with the site of oxidative phosphorylation.

4. Cyanide blocks complex IV, the last complex of the electron transport chain. What is the most immediate effect on ATP synthase?

- A) It stops making ATP because the H^+ gradient across the inner membrane dissipates.
- B) It continues making ATP because it phosphorylates ADP by substrate-level phosphorylation.
- C) It makes more ATP because protons accumulate in the matrix.
- D) It keeps working normally because ATP synthase is not part of the electron transport chain.
- E) It switches to using electrons taken directly from glucose.

Answer: A

Step 1: cyanide blocks complex IV, so electrons cannot pass to O_2 and the chain backs up. Step 2: proton pumping into the intermembrane space stops, so the H^+ gradient across the inner membrane dissipates. Step 3: ATP synthase is not itself an electron transport complex, but it depends entirely on that gradient, so ATP synthesis halts. The claim that it keeps working normally ignores this dependency, and protons would not accumulate in the matrix to boost output, because the gradient runs the other way.

5. During the complete aerobic oxidation of one glucose molecule, carbon dioxide is released:

- A) during glycolysis only
- B) during pyruvate oxidation and the citric acid cycle
- C) during glycolysis and the electron transport chain
- D) during substrate-level phosphorylation only
- E) during the electron transport chain only

Answer: B

CO_2 leaves whenever carbon exits the fuel: two molecules are released in pyruvate oxidation (the link reaction) and four more in the citric acid cycle, six per glucose. Glycolysis splits glucose into two pyruvates with no carbon lost, and the electron transport chain shuttles electrons, not carbon, so it releases none. Locating CO_2 release in the electron transport chain is therefore impossible.

6. A working muscle cell temporarily carries out only glycolysis followed by lactic acid fermentation. How many molecules of CO_2 are released per molecule of glucose consumed?

- A) 0 B) 1 C) 2 D) 3 E) 6

Answer: A

Glycolysis converts glucose to two pyruvates with no carbon lost, and lactic acid fermentation then reduces pyruvate straight to lactate with no decarboxylation step. So zero CO_2 molecules are released. The tempting count of two wrongly imports the CO_2 output of pyruvate oxidation, a step that does not occur in this anaerobic pathway.

7. Most of the carbon dioxide carried in human blood is transported:

- A) as carbonic acid bound to albumin
- B) dissolved as CO_2 in the plasma
- C) as carbaminohemoglobin bound to hemoglobin
- D) attached to the heme iron of hemoglobin
- E) as bicarbonate ions (HCO_3^-) in the plasma

Answer: E

About 70% of blood CO_2 travels as bicarbonate ions: carbonic anhydrase in red blood cells converts CO_2 and water to carbonic acid, which dissociates to HCO_3^- that then diffuses into the plasma. Carbaminohemoglobin accounts for only about 23% and about 7% stays dissolved. Also, CO_2 binds the globin protein part of hemoglobin, while the heme iron is reserved for O_2 .

8. Approximately what percentage of the carbon dioxide in human blood is transported as bicarbonate ions (HCO_3^-)?

- A) about 23% B) about 7% C) about 97% D) about 70% E) about 50%

Answer: D

Roughly 70% of CO_2 is carried as bicarbonate, formed by carbonic anhydrase in red blood cells. The 23% figure belongs to carbaminohemoglobin and 7% to CO_2 dissolved in plasma, so those two answers come from swapping the genuine transport pools, and 97% is the familiar O_2 saturation of hemoglobin, not a CO_2 statistic.

9. When a cell's ATP concentration is high, the rate of glycolysis falls. This happens because ATP:

- A) competes with fructose-6-phosphate for the active site of phosphofructokinase-1
B) activates phosphofructokinase-1 and accelerates glucose breakdown
C) binds allosterically to phosphofructokinase-1 and lowers its activity
D) competitively inhibits pyruvate kinase
E) irreversibly blocks the active site of hexokinase

Answer: C

High ATP binds an allosteric regulatory site on phosphofructokinase-1, the rate-limiting enzyme of glycolysis, and lowers its activity. It is not competitive inhibition: ATP binds away from the active site, so it does not compete with fructose-6-phosphate for the substrate pocket. Saying ATP activates PFK-1 gets the direction of regulation backwards, since AMP is the activator.

10. In a eukaryotic cell, complete oxidation of one glucose molecule yields roughly 30-32 ATP. Most of this ATP is produced by:

- A) lactic acid fermentation inside the mitochondria
B) oxidative phosphorylation powered by the electron transport chain
C) substrate-level phosphorylation in the citric acid cycle
D) direct phosphorylation during pyruvate oxidation
E) substrate-level phosphorylation in glycolysis

Answer: B

Substrate-level phosphorylation delivers only 2 ATP from glycolysis plus 2 GTP from the citric acid cycle, about 4 of the 30-32 total. The remaining roughly 26-28 ATP come from oxidative phosphorylation: NADH and FADH_2 supply electrons to the transport chain, which pumps protons to power ATP synthase. Picking any substrate-level step mistakes the small direct yield for the bulk of the total.

11. During skeletal muscle contraction, the myosin heads attach directly to:

- A) calcium ions B) the Z disc C) actin D) troponin E) tropomyosin

Answer: C

In cross-bridge cycling the myosin heads bind directly to actin and pivot to pull the thin filaments. Calcium binds troponin, which shifts tropomyosin off the myosin-binding sites on actin, so the regulatory proteins act only indirectly. Myosin never attaches to troponin, tropomyosin or calcium during contraction.

12. When oxygen is unavailable, yeast and muscle cells run fermentation chiefly in order to:

- A)** produce NADH for the electron transport chain
- B)** convert NAD⁺ into NADP⁺
- C)** remove excess NAD⁺ from the cytosol
- D)** generate extra ATP beyond that made in glycolysis
- E)** regenerate NAD⁺ so that glycolysis can continue

Answer: E

Fermentation's job is to oxidise the NADH from glycolysis back to NAD⁺ (pyruvate to lactate in muscle, to ethanol and CO₂ in yeast). Without recycled NAD⁺, glycolysis would stall once its supply ran out and the net 2 ATP per glucose would stop. The fermentation steps themselves generate no ATP, so extra ATP beyond glycolysis is a mirage.

Section B - Body Systems (Q13-20)

13. Bile, which is produced by the liver and stored in the gallbladder, assists fat digestion primarily by...

- A)** emulsifying large fat globules into tiny droplets, increasing the surface area available to lipase
- B)** hydrolysing triglycerides into fatty acids and glycerol
- C)** lowering the pH of chyme so that pancreatic enzymes can act
- D)** secreting lipase, the enzyme that chemically digests fats
- E)** activating pepsinogen into pepsin in the stomach

Answer: A

Bile contains no enzymes. Its bile salts emulsify large fat globules into tiny droplets, greatly increasing the surface area that pancreatic lipase can attack. Hydrolysing triglycerides is lipase's chemical job, and converting pepsinogen to pepsin is what stomach acid does, not bile.

14. During a normal inhalation, which sequence of events occurs?

- A)** The diaphragm relaxes and domes upward, thoracic volume decreases, and pressure inside the lungs rises above atmospheric.
- B)** The diaphragm contracts and flattens, thoracic volume increases, and pressure inside the lungs rises above atmospheric.
- C)** The diaphragm relaxes and domes upward, thoracic volume increases, and pressure in the lungs falls below atmospheric.
- D)** The diaphragm contracts and flattens, thoracic volume increases, and pressure in the lungs falls below atmospheric, drawing air in.
- E)** The diaphragm contracts and flattens, thoracic volume decreases, and pressure inside the lungs rises.

Answer: D

Step 1: the diaphragm contracts and flattens. Step 2: the thoracic cavity enlarges, so thoracic volume increases. Step 3: the larger volume lowers the pressure inside the lungs below atmospheric, and air flows in down that pressure difference. Accounts with decreasing volume or rising lung pressure describe exhalation and contradict inhalation mechanics.

15. During quiet breathing at rest, expiration is best described as...

- A)** a passive process in which the diaphragm contracts while thoracic volume stays constant
- B)** a passive process in which the diaphragm relaxes and domes upward, reducing thoracic volume
- C)** an active process in which the abdominal muscles contract forcibly to empty the lungs
- D)** an active process in which the diaphragm contracts and flattens to push air out
- E)** a passive process in which the diaphragm relaxes while thoracic volume increases

Answer: B

Quiet expiration is passive: the diaphragm relaxes and domes upward, thoracic volume falls, and the lungs' elastic recoil pushes air out. Contracting and flattening is the inhalation movement, and forcible abdominal muscle contraction belongs to forced expiration, not quiet breathing.

16. Oxygenated blood is delivered to the systemic circulation by which route?

- A)** From the left ventricle into the pulmonary artery
- B)** From the left ventricle into the aorta
- C)** From the right atrium into the vena cava
- D)** From the left atrium into the pulmonary vein
- E)** From the right ventricle into the aorta

Answer: B

Oxygenated blood returns from the lungs in the pulmonary veins to the left atrium, moves to the left ventricle, and is pumped into the aorta to supply the body. The right ventricle instead sends deoxygenated blood through the pulmonary artery to the lungs. Any route through the right side of the heart or the pulmonary artery describes pulmonary, not systemic, delivery.

17. The semilunar valves of the heart prevent...

- A)** blood from flowing back from the atria into the venae cavae and pulmonary veins
- B)** blood from flowing back from the ventricles into the atria
- C)** blood from flowing from the atria into the ventricles during diastole
- D)** blood from flowing back from the aorta and pulmonary artery into the ventricles
- E)** blood from flowing from the pulmonary veins into the left atrium

Answer: D

The semilunar valves guard the ventricular outlets in the aorta and pulmonary artery and snap shut as the ventricles relax, stopping ejected blood from falling back into them. Preventing backflow from ventricles into atria is the job of the atrioventricular valves (tricuspid and mitral), a different valve set.

18. The urinary bladder must tolerate repeated stretching as it fills with urine. Which epithelium lines it?

- A)** Stratified squamous epithelium
- B)** Simple squamous epithelium
- C)** Simple columnar epithelium
- D)** Ciliated pseudostratified epithelium
- E)** Transitional epithelium

Answer: E

Transitional epithelium (urothelium) lines the bladder and ureters: its rounded surface cells flatten as the organ fills, allowing repeated stretching and recoil without leakage. Simple columnar epithelium is built for secretion and absorption and cannot survive that distension, and stratified squamous is the abrasion-resistant lining of skin and oesophagus, not a stretchy one.

19. The kidney tubules and many secretory glands are lined mainly by which epithelium?

- A) Simple columnar epithelium
- B) Simple squamous epithelium
- C) Simple cuboidal epithelium
- D) Stratified squamous epithelium
- E) Transitional epithelium

Answer: C

Simple cuboidal epithelium lines kidney tubules and forms much of secretory glands, since its cells carry enough cytoplasm and organelles for active secretion and absorption. Simple squamous cells are paper-thin, specialised for diffusion in alveoli and vessel linings, not secretory work. Simple columnar instead dominates absorptive surfaces like the gut lining.

20. Ciliated epithelium lining the trachea serves mainly to...

- A) sweep mucus with trapped dust and microbes upward toward the pharynx
- B) contract rhythmically to narrow the airway during coughing
- C) secrete the mucus that traps inhaled particles
- D) absorb oxygen directly from the air passing through the trachea
- E) provide the main surface for gas exchange with the blood

Answer: A

The coordinated beating of the cilia sweeps the mucus blanket, with its trapped dust and microbes, upward toward the pharynx to be swallowed or expelled. The mucus is secreted by goblet cells rather than made by the cilia, and gas exchange occurs in the thin-walled alveoli, not the trachea.

Section C - Cell Division & Cell Biology (Q21-30)

21. Nondisjunction in meiosis I, meaning the failure of homologous chromosomes to separate from each other, occurs during:

- A) telophase I B) prophase I C) anaphase I D) metaphase I E) anaphase II

Answer: C

Homologous chromosomes are physically pulled to opposite poles in anaphase I, so a failure of that separation is anaphase I nondisjunction. By telophase I the chromosomes have already reached the poles, so blaming telophase comes too late. Anaphase II separates sister chromatids instead, which is the meiosis-II version of the error.

22. In a human primary spermatocyte ($2n = 46$) undergoing meiosis II, nondisjunction would be the failure of which structures to separate?

- A) chromosomes from the spindle poles
- B) bivalents at the metaphase plate
- C) homologous chromosomes of each pair
- D) sister chromatids of each chromosome
- E) non-sister chromatids at the chiasmata

Answer: D

Meiosis II separates sister chromatids, the identical copies of each chromosome joined at the centromere, so their failure to part defines meiosis-II nondisjunction. Homologous chromosomes and bivalents already separated in anaphase I, so they cannot be the structures failing in this division.

23. Independent assortment of maternal and paternal chromosomes results from their random orientation at the metaphase plate. This alignment of homologous pairs occurs during:

- A)** prophase I of meiosis
- B)** metaphase I of meiosis
- C)** metaphase of mitosis
- D)** metaphase II of meiosis
- E)** telophase I of meiosis

Answer: B

At metaphase I each homologous pair (tetrad) orients randomly relative to the spindle poles, and that orientation alone decides which maternal or paternal copy reaches each gamete, giving 2^{23} combinations in humans. By metaphase II the homologues have already separated, and only individual chromosomes align. Prophase I is for pairing and crossing over, not this alignment.

24. Crossing over, the exchange of corresponding segments between homologous chromosomes, takes place during:

- A)** prophase I **B)** anaphase I **C)** metaphase I **D)** prophase II **E)** telophase II

Answer: A

Crossing over occurs in prophase I while homologues are paired as bivalents, when chiasmata form and non-sister chromatids exchange corresponding segments. Metaphase I merely aligns the already-exchanged pairs on the spindle, so it displays the results of crossing over rather than performing it.

25. Glucose and galactose differ in the spatial arrangement of the hydroxyl group at a single chiral carbon, C-4. They are therefore classified as:

- A)** cis-trans isomers
- B)** structural isomers
- C)** enantiomers
- D)** anomers
- E)** epimers

Answer: E

Sugars differing in configuration at exactly one of several chiral centres are epimers, so glucose and galactose are C-4 epimers. Enantiomers must be mirror images at every chiral centre, which these are not, and anomers differ at the anomeric carbon (C-1) of the cyclic form, not C-4.

26. Tay-Sachs disease is caused by a defective enzyme, hexosaminidase A, so an undegraded membrane lipid (GM2 ganglioside) accumulates. In which organelle does the lipid build up?

- A)** smooth endoplasmic reticulum
- B)** peroxisome
- C)** lysosome
- D)** Golgi apparatus
- E)** mitochondrion

Answer: C

Hexosaminidase A normally degrades GM2 ganglioside inside lysosomes, the organelle holding the cell's hydrolytic enzymes, so the undigested lipid accumulates there: Tay-Sachs is a classic lysosomal storage disease. Peroxisomes instead handle very-long-chain fatty acid oxidation and catalase reactions, not ganglioside breakdown.

27. The accumulation of misfolded proteins inside the endoplasmic reticulum triggers the unfolded protein response. This misfolded-protein build-up is sensed in the:

- A)** smooth endoplasmic reticulum
- B)** cytosol
- C)** Golgi apparatus
- D)** rough endoplasmic reticulum
- E)** lysosome

Answer: D

Secretory and membrane proteins are made and folded inside the rough ER, so misfolded chains accumulate and are sensed there, triggering the unfolded protein response that slows translation and expands ER folding capacity. The smooth ER specialises in lipid synthesis and detoxification and does not fold proteins, and the Golgi only processes proteins that already folded correctly.

28. Parathyroid hormone raises blood calcium concentration. Parathyroid hormone does all of the following EXCEPT:

- A)** stimulate the release of calcitonin from the thyroid gland
- B)** increase reabsorption of calcium in the kidney
- C)** stimulate osteoclast-mediated bone resorption
- D)** increase phosphate excretion in the urine
- E)** promote activation of vitamin D

Answer: A

PTH raises blood calcium by stimulating osteoclast-mediated bone resorption, increasing renal calcium reabsorption, promoting vitamin D activation (which also raises intestinal absorption) and increasing urinary phosphate excretion. Triggering calcitonin release is the exception: calcitonin is a separate thyroid hormone that lowers blood calcium, so attributing it to PTH reverses the two hormones' opposite effects.

29. An intravenous drip delivers 0.9% saline, whose osmolarity matches that of blood plasma. Red blood cells suspended in this solution will:

- A)** shrink because the solution is hypotonic
- B)** swell because the solution is hypertonic
- C)** shrink and crenate because the solution is hypertonic
- D)** swell and possibly lyse because the solution is hypotonic
- E)** keep their normal shape because the solution is isotonic

Answer: E

0.9% saline has the same osmolarity as plasma, making it isotonic: water moves into and out of the red cells at equal rates and they keep their normal shape. Predicting swelling (hypotonic) or shrinkage (hypertonic) contradicts the stated equality of osmolarity.

30. A catastrophe reduces a large population to a few randomly surviving individuals, which then rebuild the population. The restored population shows greatly reduced genetic diversity. This is an example of:

- A)** sexual selection
- B)** the bottleneck effect
- C)** the founder effect
- D)** gene flow
- E)** stabilising selection

Answer: B

A crash to a few randomly surviving individuals that then rebuild the population is the bottleneck effect, genetic drift that strips allelic diversity. The founder effect instead describes a few individuals leaving to colonise new habitat and start a new population elsewhere; here the population rebounds in place, so the bottleneck fits.

Section D - Quantitative Chemistry (Q31-44)

31. Sulfur forms sulfuric acid, H_2SO_4 . Which of the following correctly gives both the oxidation number of sulfur in H_2SO_4 and the position of sulfur in the periodic table?

- A)** -2 ; Period 4, Group 16
- B)** $+2$; Period 2, Group 16
- C)** $+4$; Period 3, Group 16
- D)** $+6$; Period 3, Group 17
- E)** $+6$; Period 3, Group 16

Answer: E

Given: H_2SO_4 is a neutral molecule; H is $+1$, O is -2 .

Relationship: oxidation numbers in a neutral compound sum to 0: $2(+1) + \text{S} + 4(-2) = 0$.

Substitute: $2 + \text{S} + (-8) = 0$

$\text{S} - 6 = 0$

$\text{S} = +6$

Position: sulfur is in Period 3, Group 16, directly below oxygen.

Result: oxidation number $+6$; Period 3, Group 16.

Classic slip: pairing $+6$ with Period 3 Group 17 moves sulfur into chlorine's halogen slot; Group 16 sits one column to the left.

32. What is the oxidation number of manganese in KMnO_4 ?

- A)** $+2$ **B)** $+4$ **C)** $+6$ **D)** $+7$ **E)** $+8$

Answer: D

Given: KMnO_4 is a neutral compound; K is $+1$, O is -2 .

Relationship: oxidation numbers in a neutral compound sum to 0: $(+1) + \text{Mn} + 4(-2) = 0$.

Substitute: $+1 + \text{Mn} + (-8) = 0$

$\text{Mn} - 7 = 0$

$\text{Mn} = +7$

Result: $+7$.

Classic slip: $+8$ balances the four oxygens alone and drops potassium's $+1$ from the sum, and $+2$ is only the state of the Mn^{2+} ion produced when permanganate is reduced.

33. Zinc metal is placed in copper(II) sulfate solution: $\text{Zn} + \text{CuSO}_4 \rightarrow \text{ZnSO}_4 + \text{Cu}$. Which statement about this reaction is correct?

- A)** Cu metal acts as the reducing agent
- B)** Zn^{2+} is reduced because it loses electrons
- C)** Cu^{2+} is reduced because it gains electrons
- D)** Zn is reduced because it gains electrons
- E)** Cu^{2+} is oxidised because it loses electrons

Answer: C

Given: $\text{Zn} + \text{CuSO}_4 \rightarrow \text{ZnSO}_4 + \text{Cu}$; zinc starts as metal (0), copper starts as Cu^{2+} (+2).

Relationship: track oxidation numbers; losing electrons is oxidation, gaining electrons is reduction, and the reducing agent is the species that is oxidised.

Step 1: Zn goes $0 \rightarrow +2$, so zinc loses 2 electrons: oxidised, hence the reducing agent.

Step 2: Cu^{2+} goes $+2 \rightarrow 0$, so copper gains 2 electrons: reduced.

Result: Cu^{2+} is reduced because it gains electrons.

Classic slip: calling zinc 'reduced because it gains electrons' inverts the electron bookkeeping, since electron loss defines oxidation.

34. What is the hydrogen ion concentration in a solution of pH 4.0?

- A)** $1.0 \times 10^{-10} \text{ mol dm}^{-3}$
- B)** $1.0 \times 10^{-3} \text{ mol dm}^{-3}$
- C)** $1.0 \times 10^4 \text{ mol dm}^{-3}$
- D)** $1.0 \times 10^{-4} \text{ mol dm}^{-3}$
- E)** $4.0 \times 10^{-1} \text{ mol dm}^{-3}$

Answer: D

Given: $\text{pH} = 4.0$

Relationship: $\text{pH} = -\log_{10}[\text{H}^+]$, so the concentration is 10 raised to minus the pH.

Substitute: $[\text{H}^+] = 10^{-4} \text{ mol dm}^{-3}$

$10^{-4} = 1.0 \times 10^{-4} \text{ mol dm}^{-3}$

Result: $1.0 \times 10^{-4} \text{ mol dm}^{-3}$.

Classic slip: $1.0 \times 10^{-10} \text{ mol dm}^{-3}$ is the hydroxide concentration instead, since $\text{pOH} = 14 - 4 = 10$.

35. At 25°C , $K_w = 1.0 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$. A solution has $[\text{OH}^-] = 1.0 \times 10^{-4} \text{ mol dm}^{-3}$. What is the pH of this solution?

- A)** 4 **B)** 7 **C)** 10 **D)** 11 **E)** 14

Answer: C

Given: $[\text{OH}^-] = 1.0 \times 10^{-4} \text{ mol dm}^{-3}$ at 25°C , where $K_w = 1.0 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$

Relationship: $\text{pOH} = -\log_{10}[\text{OH}^-]$, and $\text{pH} + \text{pOH} = 14$ because $K_w = [\text{H}^+][\text{OH}^-] = 10^{-14}$.

Substitute: $\text{pOH} = -\log_{10}(1.0 \times 10^{-4}) = 4$

$\text{pH} = 14 - 4$

$\text{pH} = 10$

Result: $\text{pH} = 10$

Classic slip: answering 4 reports the pOH itself, and 7 would need the neutral $[\text{OH}^-]$ of $10^{-7} \text{ mol dm}^{-3}$.

36. What is the pH of a $0.001 \text{ mol dm}^{-3}$ solution of hydrochloric acid, a strong monoprotic acid?

- A)** -3 **B)** 0.001 **C)** 2 **D)** 3 **E)** 11

Answer: D

Given: HCl at $0.001 \text{ mol dm}^{-3}$, a strong monoprotic acid.

Relationship: a strong acid dissociates completely, so $[\text{H}^+]$ equals the acid concentration; $\text{pH} = -\log_{10}[\text{H}^+]$.

Substitute: $[\text{H}^+] = 0.001 \text{ mol dm}^{-3} = 1 \times 10^{-3} \text{ mol dm}^{-3}$

$$\text{pH} = -\log_{10}(10^{-3})$$

$$\text{pH} = 3$$

Result: $\text{pH} = 3$

Classic slip: 11 equals $14 - 3$ and is the pOH of this solution, while -3 is the logarithm with its sign dropped.

37. How many molecules are present in 22.4 dm^3 of oxygen gas at STP? (Molar volume at STP = $22.4 \text{ dm}^3 \text{ mol}^{-1}$; Avogadro constant = $6.02 \times 10^{23} \text{ mol}^{-1}$)

- A)** 1.204×10^{24} **B)** 1.35×10^{25} **C)** 3.01×10^{23} **D)** 6.02×10^{23} **E)** 6.02×10^{22}

Answer: D

Given: $V = 22.4 \text{ dm}^3$ of O_2 at STP; molar volume = $22.4 \text{ dm}^3 \text{ mol}^{-1}$; Avogadro constant = $6.02 \times 10^{23} \text{ mol}^{-1}$.

Relationship: moles $n = V \div \text{molar volume}$; number of molecules = $n \times \text{Avogadro constant}$.

Substitute: $n = 22.4 \text{ dm}^3 \div 22.4 \text{ dm}^3 \text{ mol}^{-1} = 1 \text{ mol}$

$$\text{molecules} = 1 \text{ mol} \times 6.02 \times 10^{23} \text{ mol}^{-1}$$

Result: 6.02×10^{23} molecules.

Classic slip: doubling to 1.204×10^{24} assumes 2 mol of gas, and 1.35×10^{25} comes from multiplying 6.02×10^{23} by 22.4 as if the molar volume were a mole count.

38. 0.5 mol of sodium hydroxide is dissolved in water and the solution is made up to 250 cm^3 . What is the concentration of the solution in mol dm^{-3} ?

- A)** $0.125 \text{ mol dm}^{-3}$ **B)** 0.5 mol dm^{-3} **C)** 1.25 mol dm^{-3} **D)** 2.0 mol dm^{-3} **E)** 5.0 mol dm^{-3}

Answer: D

Given: $n = 0.5 \text{ mol}$ of NaOH; final solution volume 250 cm^3 .

Relationship: concentration $c = n \div V$, with V converted to dm^3 ($1 \text{ dm}^3 = 1000 \text{ cm}^3$)

Substitute: $V = 250 \text{ cm}^3 \div 1000 = 0.250 \text{ dm}^3$

$$c = 0.5 \text{ mol} \div 0.250 \text{ dm}^3$$

$$c = 2.0 \text{ mol dm}^{-3}$$

Result: 2.0 mol dm^{-3} .

Classic slip: $0.125 \text{ mol dm}^{-3}$ multiplies by the volume (0.5×0.25) instead of dividing by it, and 0.5 mol dm^{-3} ignores the made-up-to- 250 cm^3 dilution.

39. Which law states that the volume of a gas is directly proportional to the number of moles of gas present, provided the temperature and pressure are constant?

- A)** Avogadro's law
- B)** Boyle's law
- C)** Charles's law
- D)** Gay-Lussac's law
- E)** Dalton's law of partial pressures

Answer: A

Volume proportional to number of moles at constant temperature and pressure is Avogadro's law: equal volumes of gas at the same T and P contain equal moles. Charles's law links volume to absolute temperature, Boyle's law links volume to pressure, and Dalton's law deals with partial pressures, so none of those describes a moles-volume relationship.

40. A gas occupies 9.0 dm^3 at 27°C and constant pressure. It is then cooled to -73°C , still at the same pressure. What volume does it now occupy?

- A)** -24.3 dm^3 **B)** 3.0 dm^3 **C)** 6.0 dm^3 **D)** 13.5 dm^3 **E)** 24.3 dm^3

Answer: C

Given: $V_1 = 9.0 \text{ dm}^3$ at $T_1 = 27^\circ\text{C}$; cooled to $T_2 = -73^\circ\text{C}$; pressure constant

Relationship: Charles's law, $V_1/T_1 = V_2/T_2$, with all temperatures in kelvin ($K = ^\circ\text{C} + 273$).

Substitute: $T_1 = 27 + 273 = 300 \text{ K}$; $T_2 = -73 + 273 = 200 \text{ K}$

$$V_2 = 9.0 \text{ dm}^3 \times (200 \div 300)$$

$$200 \div 300 = 2/3, \text{ so } V_2 = 9.0 \times 2/3 \text{ dm}^3$$

$$V_2 = 6.0 \text{ dm}^3$$

Result: 6.0 dm^3 .

Classic slip: inserting the Celsius readings directly gives a negative volume (-24.3 dm^3), impossible for a gas, and any expansion contradicts cooling at constant pressure.

41. A gas occupies 500 cm^3 at a pressure of 100 kPa , the temperature remaining constant. The pressure is increased to 250 kPa . What is the new volume?

- A)** 50 cm^3 **B)** 100 cm^3 **C)** 200 cm^3 **D)** 250 cm^3 **E)** 1250 cm^3

Answer: C

Given: $V_1 = 500 \text{ cm}^3$ at $P_1 = 100 \text{ kPa}$; $P_2 = 250 \text{ kPa}$; temperature constant.

Relationship: Boyle's law, $P_1V_1 = P_2V_2$, so $V_2 = P_1 \times V_1 \div P_2$

Substitute: $V_2 = 100 \text{ kPa} \times 500 \text{ cm}^3 \div 250 \text{ kPa}$

$$100 \times 500 = 50,000$$

$$50,000 \div 250 = 200 \text{ cm}^3$$

Result: 200 cm^3 .

Classic slip: 1250 cm^3 multiplies by the inverted pressure ratio ($250 \div 100$) even though raising the pressure must compress the gas, never expand it.

42. Each of the following compounds contains two carbon atoms. Which has the highest boiling point?

- A) CH_3CH_3 (ethane)
- B) CH_3COOH (ethanoic acid)
- C) CH_3OCH_3 (methoxymethane)
- D) CH_3CHO (ethanal)
- E) $\text{CH}_3\text{CH}_2\text{OH}$ (ethanol)

Answer: B

Boiling point tracks intermolecular force strength. Ethanoic acid molecules form cyclic dimers held by two hydrogen bonds at once, giving about 118 C, well above ethanol's one-to-one hydrogen bonding at about 78 C. Ethanal, methoxymethane and ethane lack an O-H hydrogen-bond donor, so dipole and then dispersion forces make them progressively lower-boiling.

43. Which one of the following molecules is polar?

- A) NH_3
- B) BF_3
- C) CO_2
- D) CCl_4
- E) BeCl_2

Answer: A

NH_3 is pyramidal with a lone pair on nitrogen, so its three N-H bond dipoles point asymmetrically and do not cancel: it is polar. CO_2 and BeCl_2 (linear), BF_3 (trigonal planar) and CCl_4 (tetrahedral) are all symmetric, so their bond dipoles cancel and the molecules are non-polar despite having polar bonds.

44. Given the following data: $\text{C(s)} + \text{O}_2\text{(g)} \rightarrow \text{CO}_2\text{(g)}$, $\Delta H = -394 \text{ kJ mol}^{-1}$; $\text{CO(g)} + \frac{1}{2}\text{O}_2\text{(g)} \rightarrow \text{CO}_2\text{(g)}$, $\Delta H = -283 \text{ kJ mol}^{-1}$. What is ΔH for the reaction $\text{C(s)} + \frac{1}{2}\text{O}_2\text{(g)} \rightarrow \text{CO(g)}$?

- A) -677 kJ mol^{-1}
- B) -394 kJ mol^{-1}
- C) -283 kJ mol^{-1}
- D) -111 kJ mol^{-1}
- E) $+111 \text{ kJ mol}^{-1}$

Answer: D

Given: $\text{C(s)} + \text{O}_2\text{(g)} \rightarrow \text{CO}_2\text{(g)}$ with $\Delta H = -394 \text{ kJ mol}^{-1}$; $\text{CO(g)} + \frac{1}{2}\text{O}_2\text{(g)} \rightarrow \text{CO}_2\text{(g)}$ with $\Delta H = -283 \text{ kJ mol}^{-1}$

Target: ΔH for $\text{C(s)} + \frac{1}{2}\text{O}_2\text{(g)} \rightarrow \text{CO(g)}$.

Relationship: Hess's law; reversing an equation flips the sign of its ΔH , and combined equations add their ΔH values.

Step 1: reverse the second equation: $\text{CO}_2\text{(g)} \rightarrow \text{CO(g)} + \frac{1}{2}\text{O}_2\text{(g)}$, $\Delta H = +283 \text{ kJ mol}^{-1}$

Step 2: add it to the first equation; CO_2 cancels and $1 \text{ O}_2 - \frac{1}{2} \text{ O}_2 = \frac{1}{2} \text{ O}_2$ remains as reactant.

Step 3: the net equation is $\text{C(s)} + \frac{1}{2}\text{O}_2\text{(g)} \rightarrow \text{CO(g)}$, so add the enthalpies: $-394 + 283$.

$\Delta H = -111 \text{ kJ mol}^{-1}$

Result: -111 kJ mol^{-1} .

Classic slip: -677 kJ mol^{-1} adds both given values without reversing the second equation and flipping its sign.

Section E - Mathematics (Q45-53)

45. Solve for x : $\sqrt{3x - 2} = 4$.

- A)** $x = 18$ **B)** $x = 14/3$ **C)** $x = 4$ **D)** $x = 2$ **E)** $x = 6$

Answer: E

Given: $\sqrt{3x - 2} = 4$

Relationship: square both sides to remove the root (both sides are non-negative, so nothing is lost), then solve the linear equation.

Substitute: $(\sqrt{3x - 2})^2 = 4^2$

$$3x - 2 = 16$$

$$3x = 16 + 2 = 18$$

$$x = 18 \div 3 = 6$$

Check: $\sqrt{3 \times 6 - 2} = \sqrt{16} = 4$, valid

Result: $x = 6$

Classic slip: $x = 14/3$ comes from writing $3x = 16 - 2$, subtracting the 2 after squaring instead of adding it.

46. Solve for x : $\sqrt{x + 6} = x$. Which answer lists ALL valid solutions?

- A)** $x = 6$
B) $x = 3$ only
C) $x = -2$ only
D) $x = 3$ and $x = -2$
E) $x = 2$

Answer: B

Given: $\sqrt{x + 6} = x$

Relationship: square both sides, solve the quadratic, then substitute every candidate back, because squaring can manufacture roots the original rejects.

Substitute: $x + 6 = x^2$

$$\text{Rearrange: } x^2 - x - 6 = 0$$

$$\text{Factorise: } (x - 3)(x + 2) = 0, \text{ so } x = 3 \text{ or } x = -2$$

Check $x = 3$: $\sqrt{3 + 6} = \sqrt{9} = 3$, valid

Check $x = -2$: $\sqrt{-2 + 6} = \sqrt{4} = 2$, never -2 , so it is extraneous

Result: $x = 3$ only

Classic slip: reporting both candidates forgets that a square root can never equal a negative number.

47. Solve the inequality: $5 - 3x \geq 20$.

- A) $x \geq -5$ B) $x \leq -5$ C) $x \geq 5$ D) $x < -5$ E) $x \leq 5$

Answer: B

Given: $5 - 3x \geq 20$.

Relationship: isolate x ; dividing an inequality by a negative number flips its direction.

Step 1: subtract 5: $-3x \geq 20 - 5$.

$$-3x \geq 15.$$

Step 2: divide both sides by -3 and flip $\geq$ to $\leq$.

$$x \leq 15 \div (-3).$$

$$x \leq -5.$$

Result: $x \leq -5$.

Classic slip: $x \geq -5$ keeps the old direction and skips the flip, and a strict $x < -5$ would drop the inclusive boundary the original $\geq$ carries.

48. Solve the inequality: $-2x + 7 > 15$.

- A) $x < -4$ B) $x < 4$ C) $x > -4$ D) $x > 4$ E) $x < -11$

Answer: A

Given: $-2x + 7 > 15$.

Relationship: isolate x ; dividing by a negative number flips the inequality.

Step 1: subtract 7: $-2x > 15 - 7$.

$$-2x > 8.$$

Step 2: divide both sides by -2 and flip $>$ to $<$.

$$x < 8 \div (-2).$$

$$x < -4.$$

Result: $x < -4$.

Classic slip: $x > -4$ forgets the flip, and $x < -11$ comes from moving the 7 the wrong way, giving $-2x > 22$.

49. If $f(x) = 3x^2 - 4x + 5$, what is $f(2)$?

- A) 4 B) 9 C) 13 D) 25 E) 33

Answer: B

Given: $f(x) = 3x^2 - 4x + 5$, evaluated at $x = 2$

Relationship: substitute into each term, squaring before multiplying.

$$\text{Substitute: } f(2) = 3 \times 2^2 - 4 \times 2 + 5$$

$$2^2 = 4, \text{ so the first term is } 3 \times 4 = 12$$

$$\text{The second term is } 4 \times 2 = 8$$

$$f(2) = 12 - 8 + 5$$

$$f(2) = 4 + 5 = 9$$

Result: $f(2) = 9$

Classic slip: 13 treats $-4x$ as a bare -4 after substituting, and 33 squares the coefficient too, as $(3 \times 2)^2 = 36$ instead of $3 \times 2^2 = 12$.

50. A 15 m ladder leans against a vertical wall with its foot 9 m from the base of the wall. How high up the wall does the ladder reach?

- A)** 6 m **B)** 12 m **C)** 13 m **D)** 17.5 m **E)** 24 m

Answer: B

Given: ladder 15 m (hypotenuse), foot 9 m from the wall, wall height h unknown.

Relationship: Pythagoras, $\text{hypotenuse}^2 = \text{base}^2 + \text{height}^2$, so $h = \sqrt{15^2 - 9^2}$.

Substitute: $h = \sqrt{225 - 81}$ m

$h = \sqrt{144}$ m

$h = 12$ m

Result: 12 m.

Classic slip: 17.5 m adds the squares, $\sqrt{225 + 81}$, treating the wall as the hypotenuse when the ladder is; the scaled 3-4-5 triple 9-12-15 confirms 12 m.

51. A rectangle has a width of 16 cm and a diagonal of 34 cm. What is its perimeter?

- A)** 46 cm **B)** 62 cm **C)** 92 cm **D)** 100 cm **E)** 128 cm

Answer: C

Given: rectangle width 16 cm, diagonal 34 cm.

Relationship: the width, height and diagonal form a right triangle: $\text{height} = \sqrt{34^2 - 16^2}$; $\text{perimeter} = 2 \times (\text{width} + \text{height})$.

Substitute: $34^2 = 1156$ and $16^2 = 256$

$\text{height} = \sqrt{1156 - 256}$ cm

$\text{height} = \sqrt{900}$ cm = 30 cm

$\text{perimeter} = 2 \times (16 + 30)$ cm = 2×46 cm

$\text{perimeter} = 92$ cm

Result: 92 cm.

Classic slip: 100 cm treats the diagonal as the second side, $2 \times (16 + 34)$, and 46 cm forgets to double the half-perimeter.

52. Solve for x : $4(x - 2) = 2x + 2$.

- A)** $x = 4$ **B)** $x = -3$ **C)** $x = 2$ **D)** $x = 10$ **E)** $x = 5$

Answer: E

Given: $4(x - 2) = 2x + 2$

Relationship: expand the bracket so the 4 multiplies both terms inside, then collect x on one side.

Substitute (expand): $4x - 8 = 2x + 2$

Subtract $2x$ from both sides: $2x - 8 = 2$

Add 8 to both sides: $2x = 10$

$x = 10 \div 2 = 5$

Check: $4(5 - 2) = 12$ and $2 \times 5 + 2 = 12$

Result: $x = 5$

Classic slip: $x = 2$ comes from distributing the 4 to only the first term inside the bracket, writing $4x - 2$.

53. Using $\pi = 3.14$, what is the area of a circle with a diameter of 12 cm?

- A)** 37.68 cm² **B)** 75.36 cm² **C)** 113.04 cm² **D)** 226.08 cm² **E)** 452.16 cm²

Answer: C

Given: circle of diameter 12 cm; $\pi = 3.14$

Relationship: radius $r = \text{diameter} \div 2$; area $A = \pi r^2$

Substitute: $r = 12 \div 2 = 6$ cm

$$A = 3.14 \times 6^2 \text{ cm}^2$$

$$6^2 = 36$$

$$A = 3.14 \times 36 \text{ cm}^2$$

$$A = 113.04 \text{ cm}^2$$

Result: 113.04 cm².

Classic slip: 452.16 cm² squares the diameter instead of the radius (3.14×144), and 37.68 cm² is the circumference, $2 \times 3.14 \times 6$.

Section F - Physics (Q54-62)

54. A car travels at 72 km/h. What is this speed in m/s?

- A)** 7.2 m/s **B)** 20 m/s **C)** 24 m/s **D)** 36 m/s **E)** 240 m/s

Answer: B

Given: speed 72 km/h.

Relationship: 1 km = 1000 m and 1 h = 3600 s, so converting km/h to m/s divides by 3.6

Substitute: $72 \text{ km/h} \times 1000 \text{ m} \div 1 \text{ km} = 72,000 \text{ m per hour}$

$72,000 \text{ m} \div 3600 \text{ s}$.

$$72,000 \div 3600 = 20 \text{ m/s}$$

Result: 20 m/s.

Classic slip: 36 m/s divides by 2 and 240 m/s multiplies instead; the m/s figure is always the smaller number.

55. A cyclist rides at 15 m/s. What is this speed in km/h?

- A)** 4.2 km/h **B)** 5 km/h **C)** 45 km/h **D)** 54 km/h **E)** 150 km/h

Answer: D

Given: speed 15 m/s.

Relationship: 1 m/s covers 3600 m = 3.6 km in one hour, so converting m/s to km/h multiplies by 3.6.

Substitute: $15 \text{ m/s} \times 3.6$.

$$15 \times 3 = 45 \text{ and } 15 \times 0.6 = 9$$

$$45 + 9 = 54 \text{ km/h}$$

Result: 54 km/h.

Classic slip: 4.2 km/h divides by 3.6 instead of multiplying, even though the km/h value must come out larger.

56. Two $6\ \Omega$ resistors are connected in parallel. What is the total resistance?

- A)** $1/3\ \Omega$ **B)** $3\ \Omega$ **C)** $6\ \Omega$ **D)** $12\ \Omega$ **E)** $36\ \Omega$

Answer: B

Given: two resistors in parallel, $R_1 = 6\ \Omega$ and $R_2 = 6\ \Omega$

Relationship: for parallel resistors $1/R = 1/R_1 + 1/R_2$, then invert to get R .

Substitute: $1/R = (1/6 + 1/6)\ \Omega^{-1}$

$$1/R = 2/6\ \Omega^{-1} = 1/3\ \Omega^{-1}$$

$$R = 3\ \Omega$$

Result: $3\ \Omega$.

Classic slip: $1/3\ \Omega$ stops after summing and never inverts, and $12\ \Omega$ is the series sum even though a parallel total must sit below the smallest branch.

57. Three resistors of $4\ \Omega$, $4\ \Omega$ and $2\ \Omega$ are connected in parallel. What is the total resistance?

- A)** $0.25\ \Omega$ **B)** $1\ \Omega$ **C)** $2\ \Omega$ **D)** $3.3\ \Omega$ **E)** $10\ \Omega$

Answer: B

Given: three resistors in parallel: $4\ \Omega$, $4\ \Omega$ and $2\ \Omega$.

Relationship: $1/R = 1/R_1 + 1/R_2 + 1/R_3$, then invert to get R

Substitute: $1/R = (1/4 + 1/4 + 1/2)\ \Omega^{-1}$

$$1/R = (0.25 + 0.25 + 0.5)\ \Omega^{-1}$$

$$1/R = 1.0\ \Omega^{-1}$$

$$R = 1\ \Omega$$

Result: $1\ \Omega$.

Classic slip: $10\ \Omega$ is the plain sum, which would only fit a series connection; a parallel total must fall below the smallest branch, $2\ \Omega$.

58. A current of $3\ \text{A}$ flows through a $4\ \Omega$ resistor. What power is dissipated?

- A)** $7\ \text{W}$ **B)** $12\ \text{W}$ **C)** $36\ \text{W}$ **D)** $48\ \text{W}$ **E)** $144\ \text{W}$

Answer: C

Given: current $I = 3\ \text{A}$ through resistance $R = 4\ \Omega$

Relationship: power dissipated in a resistor, $P = I^2R$

Substitute: $P = 3^2 \times 4\ \text{W}$

$$3^2 = 9$$

$$P = 9 \times 4\ \text{W}$$

$$P = 36\ \text{W}$$

Result: $36\ \text{W}$.

Classic slip: $12\ \text{W}$ uses $P = IR$ without squaring the current, $144\ \text{W}$ squares the whole product (12^2), and $48\ \text{W}$ squares the resistance; only the current carries the square.

59. An appliance draws a current of 2.5 A when connected to a 12 V supply. What power does it use?

- A) 0.21 W B) 4.8 W C) 14.5 W D) 30 W E) 60 W

Answer: D

Given: supply voltage $V = 12 \text{ V}$, current $I = 2.5 \text{ A}$

Relationship: electrical power $P = V \times I$.

Substitute: $P = 12 \text{ V} \times 2.5 \text{ A}$

$$12 \times 2.5 = 12 \times 2 + 12 \times 0.5 = 24 + 6$$

$$P = 30 \text{ W}$$

Result: 30 W.

Classic slip: 4.8 W divides voltage by current and 14.5 W adds them; power is strictly the product of voltage and current.

60. What is the gauge pressure at a depth of 5 m in water? ($\rho = 1000 \text{ kg/m}^3$, $g = 9.8 \text{ m/s}^2$)

- A) 4,900 Pa B) 9,800 Pa C) 49,000 Pa D) 49 Pa E) 490,000 Pa

Answer: C

Given: water density $\rho = 1000 \text{ kg/m}^3$, $g = 9.8 \text{ m/s}^2$, depth $h = 5 \text{ m}$

Relationship: gauge pressure in a liquid, $P = \rho \times g \times h$.

Substitute: $P = 1000 \text{ kg/m}^3 \times 9.8 \text{ m/s}^2 \times 5 \text{ m}$

$$1000 \times 9.8 = 9,800$$

$$9,800 \times 5 = 49,000$$

$$P = 49,000 \text{ Pa}$$

Result: 49,000 Pa.

Classic slip: 9,800 Pa stops after $\rho \times g$ and never multiplies by the depth, and 4,900 Pa drops a factor of ten.

61. A block has a mass of 540 g and a volume of 200 cm^3 . What is its density, and which metal is it consistent with?

- A) 0.37 g/cm^3 — a light alloy
B) 2.7 g/cm^3 — aluminium
C) 27 g/cm^3 — a heavy alloy
D) 108 g/cm^3 — gold
E) 740 g/cm^3 — lead

Answer: B

Given: mass $m = 540 \text{ g}$, volume $V = 200 \text{ cm}^3$

Relationship: density $\rho = m \div V$.

Substitute: $\rho = 540 \text{ g} \div 200 \text{ cm}^3$

$$540 \div 200 = 2.7$$

$$\rho = 2.7 \text{ g/cm}^3$$

Result: 2.7 g/cm^3 , which matches aluminium.

Classic slip: 0.37 g/cm^3 inverts the division ($200 \div 540$), even though dividing 540 by the smaller 200 must land above 1.

62. A lamp produces a light intensity of 90 W/m^2 at a distance of 1 m. What is the intensity at 3 m from the lamp?

- A)** 10 W/m^2 **B)** 30 W/m^2 **C)** 90 W/m^2 **D)** 270 W/m^2 **E)** 810 W/m^2

Answer: A

Given: intensity 90 W/m^2 at distance 1 m; new distance 3 m.

Relationship: inverse-square law, intensity $\propto 1/\text{distance}^2$, so $I_2 = I_1 \times (d_1 \div d_2)^2$.

Substitute: $I_2 = 90 \times (1 \div 3)^2 \text{ W/m}^2$

$$(1 \div 3)^2 = 1/9$$

$$I_2 = 90 \div 9 \text{ W/m}^2$$

$$I_2 = 10 \text{ W/m}^2$$

Result: 10 W/m^2 .

Classic slip: 30 W/m^2 divides by 3 only, reading the law as inverse rather than inverse-square; tripling the distance must divide by 9.

Section G - Logic & Reasoning (Q63-70)

63. 'If Maria catches the 8:00 train, she arrives before noon.' Maria did not arrive before noon. Which conclusion necessarily follows?

- A)** If Maria does not catch the 8:00 train, she does not arrive before noon.
B) If Maria arrives before noon, she caught the 8:00 train.
C) Maria missed the 8:00 train because she overslept.
D) Maria caught the 8:00 train, but the train was delayed.
E) Maria did not catch the 8:00 train.

Answer: E

Given: P = Maria catches the 8:00 train; Q = she arrives before noon; premise $P \rightarrow Q$; observed fact: not Q .

Relationship: a conditional is equivalent to its contrapositive: $P \rightarrow Q$ means $\text{not } Q \rightarrow \text{not } P$.

Step 1: the fact is not Q (she did not arrive before noon).

Step 2: apply $\text{not } Q \rightarrow \text{not } P$.

Step 3: therefore not P : she did not catch the 8:00 train.

Result: Maria did not catch the 8:00 train.

Classic slip: the converse (arrived before noon, so she caught the train) and the inverse (missed the train, so no early arrival) are not licensed, and delay or oversleeping stories add facts that contradict the premises.

64. A technician states: 'If the sample is contaminated, the result is unreliable.' Which statement, if true, contradicts this?

- A)** The sample is contaminated and the result is reliable.
- B)** The sample is not contaminated and the result is unreliable.
- C)** The sample is contaminated and the result is unreliable.
- D)** If the sample is not contaminated, the result is reliable.
- E)** The sample is not contaminated and the result is reliable.

Answer: A

Given: conditional $P \rightarrow Q$, with P = the sample is contaminated, Q = the result is unreliable.

Relationship: a conditional is false only when its antecedent is true and its consequent is false; that one combination is its contradiction.

Step 1: force P true (contaminated sample).

Step 2: force Q false (reliable result).

Step 3: P true with Q false is exactly the failing case, so the statement is contradicted.

Result: the sample is contaminated and the result is reliable.

Classic slip: an uncontaminated sample with an unreliable result looks adverse but leaves the promise untouched, because the promise covers only contaminated samples.

65. Anna reasons: 'If the alarm rings, the neighbours call the police. The alarm did not ring, so the neighbours did not call the police.' Her reasoning is...

- A)** invalid: the original conditional itself must be false because the neighbours might call anyway
- B)** valid: it correctly applies the contrapositive of the original statement
- C)** valid: the inverse of a conditional always follows from it
- D)** valid: the neighbours can only call the police when the alarm rings
- E)** invalid: it denies the antecedent, and the inverse is not equivalent to the original conditional

Answer: E

Given: premise $P \rightarrow Q$ (alarm rings $\rightarrow$ neighbours call); observed not P (alarm silent); Anna concludes not Q .

Relationship: test the argument form: concluding not Q from not P is denying the antecedent, an invalid pattern.

Step 1: the premise constrains only cases where P holds.

Step 2: with P false the premise is silent; the neighbours may still call for another reason.

Step 3: so not Q does not follow from not P .

Result: invalid: it denies the antecedent, and the inverse is not equivalent to the original conditional.

Classic slip: the valid move is the contrapositive (no call from the neighbours implies the alarm did not ring), which runs in the opposite direction.

66. All cardiologists are doctors. Some teachers are doctors. No cardiologist is a teacher. Which conclusion follows necessarily?

- A)** All teachers are doctors.
- B)** All doctors are either cardiologists or teachers.
- C)** No teacher is a doctor.
- D)** Some cardiologists are teachers.
- E)** Some teachers are not cardiologists.

Answer: E

Given: all cardiologists are doctors; some teachers are doctors; no cardiologist is a teacher.

Relationship: combine the some-teachers-are-doctors premise with the no-overlap premise.

Step 1: at least one teacher is a doctor (second premise).

Step 2: no cardiologist is a teacher, so that doctor-teacher cannot be a cardiologist.

Step 3: such a person exists, so at least one teacher is not a cardiologist.

Result: some teachers are not cardiologists.

Classic slip: 'all teachers are doctors' inflates a some-premise into a universal, and 'some cardiologists are teachers' collides head-on with the no-overlap premise.

67. No reptiles are warm-blooded. All snakes are reptiles. Which conclusion follows necessarily?

- A)** No snakes are warm-blooded.
- B)** All warm-blooded animals are reptiles.
- C)** Some snakes are warm-blooded.
- D)** No animal other than reptiles is warm-blooded.
- E)** Some reptiles are warm-blooded.

Answer: A

Given: no reptiles are warm-blooded; all snakes are reptiles.

Relationship: chain the set inclusions: snakes sit wholly inside reptiles, and reptiles sit wholly outside the warm-blooded set.

Step 1: every snake is a reptile.

Step 2: every reptile is not warm-blooded.

Step 3: so every snake is not warm-blooded.

Result: no snakes are warm-blooded.

Classic slip: 'all warm-blooded animals are reptiles' reverses the exclusion, and any warm-blooded snake or reptile claim collides with the first premise.

68. A study of 5,000 cyclists found that riders without helmets were more than three times as likely to sustain head injuries as riders wearing helmets. Which conclusion is best supported?

- A)** In this study, unhelmeted riders had a higher incidence of head injury than helmeted riders.
- B)** Cycling without a helmet causes riders to crash more often.
- C)** Unhelmeted riders take more risks than helmeted riders.
- D)** Helmets prevent all head injuries in cycling accidents.
- E)** Helmet laws should be introduced to reduce all cycling injuries.

Answer: A

The data license only the within-study comparison: unhelmeted riders had a higher incidence of head injury than helmeted riders. Risk-taking behaviour, crash frequency, complete prevention and legislative recommendations all import causal or universal claims that observational numbers never established.

- 69.** A survey of 2,000 employees found that those who walk to work reported, on average, higher job satisfaction than those who commute by car. Which conclusion follows?
- A)** Car commuters would become more satisfied if they switched to walking.
 - B)** Employers should subsidise walking programmes in order to raise productivity.
 - C)** Walking to work causes greater job satisfaction.
 - D)** Every employee who walks to work is more satisfied than every employee who drives.
 - E)** Among surveyed employees, walking commuters reported higher average satisfaction than car commuters.

Answer: E

The survey supports exactly one comparison: among surveyed employees, walking commuters reported higher average satisfaction than car commuters. An average hides individual variation, so 'every walker more satisfied than every driver' overreaches, and the causal claim, the switching counterfactual and the productivity recommendation all go beyond the correlational data.

- 70.** What is the angle between the hour hand and the minute hand of an analogue clock at 3:15?
- A)** 0 degrees **B)** 7.5 degrees **C)** 15 degrees **D)** 30 degrees **E)** 82.5 degrees

Answer: B

Given: time 3:15; minute hand 15 minutes past 12; hour hand a quarter hour past the 3.

Relationship: the minute hand sweeps 6 degrees per minute; the hour hand sweeps 30 degrees per hour plus 0.5 degrees per minute.

Step 1: minute hand from 12 = $15 \times 6 = 90$ degrees.

Step 2: hour hand from 12 = $3 \times 30 + 15 \times 0.5$ degrees.

Step 3: $3 \times 30 = 90$ and $15 \times 0.5 = 7.5$, so the hour hand sits at 97.5 degrees.

Step 4: angle between the hands = $97.5 - 90 = 7.5$ degrees.

Result: 7.5 degrees.

Classic slip: 0 degrees assumes both hands rest on the 3, ignoring the hour hand's drift one quarter of the way toward the 4.